

sol1_enhancer_agent_sdk

Take-home. Python owns the loop. The model only drafts and grades.

Stop conditions cannot be talked past.

Needs `claude-agent-sdk` and an API key for a live poll. `task test` does not.

What you will build

FILE	JOB
enhancer.py	one poll-and-act step in Python
loop.py	CLI --once, --table-only
roles.py	ClaudeAgentOptions plus scope_hook
plugin/	same agents as Saturday, loaded with plugins=
check_fields.py	ready is arithmetic
check_stop.py	done, cost, max turns
tests/	pytest, stubs the SDK

The point is not that it runs. The point is that the rubric, the write scope, and the exits did not have to change.

Why Python holds the loop

A skill is a model following steps. A model can skip a step.

Here the parent session may only use the `Agent` tool. Python writes the candidate from the doer's text. Python calls `check_fields.py`. Python calls `check_stop.py`.

Learning objectives

- Configure `tools=[...]` so judge and doer hold no Write
- Implement `scope_hook` with the full deny envelope
- Validate `--table-only` prints judge writes = no
- Troubleshoot a hook that fails **open** because of a typo
- Name the marker gap vs Deep Agents
- Deploy `loop.py --once` on issue events

Starting architecture

```
python loop.py --once --repo TARGET --ticket T900
└─ enhancer.py
    Agent tool
    │ └─ enhancer-judge  Read, Grep, Glob  output_format JSON
    └─ enhancer-doer    Read, Grep, Glob  text draft
Python writes tickets/<id>.enhancer-candidate.md
check_fields.py / check_stop.py
gh labels + .harness/last-enhancer-<id>.json
```

Scope. Two places, both required

1. `tools=[...].NO_WRITE` strips Edit, Write, Bash.
2. `PreToolUse` `hook` `roles.scope_hook`.

Empty `{}` = allow. Deny must be:

```
{
  "hookSpecificOutput": {
    "hookEventName": "PreToolUse",
    "permissionDecision": "deny",
    "permissionDecisionReason": "...",
  }
}
```

A typo fails **open**. Tests assert this shape key by key.

Parent session

```
allowed_tools=["Agent"]
disallowed_tools=NO_WRITE
CLAUDE_AGENT_SDK_DISABLE_BUILTIN_AGENTS=1
maxTurns=12
background=False
```

System prompt: "Python already owns the loop. Do not invoke the enhancer-loop skill."

Judge uses structured `output_format(JUDGE_SCHEMA)`. The hook still fires if a Write leaks.

Cast. `roleplan.py`

```
enhancer: orchestrator, doer, judge
orchestrator  tools: Task                writes: no
doer          tools: Read,Glob,Grep>Edit,Write,Bash  allow: tickets/**
judge         tools: Read,Glob,Grep,Bash             writes: no
```

Doer tools include Write in the table. Python still writes the candidate. The hook is the fail-closed belt.

Extra exits. Marker gap

`check_stop.py` also fires on `usd >= max_usd` and `turns >= max_turns`.

Known gap vs the plugin and Deep Agents: `Gh.latest_comment` does not filter `<!-- enhancer-loop -->`, and `comment()` does not append it. Do not copy that gap into a new port.

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Commands

```
cd solutions/sol1_enhancer_agent_sdk
pip install -r ../../requirements-takehome.txt
cp config.json.example config.json
task table          # python loop.py --table-only
task test           # pytest, no key, no clone
task checks
task clone && task create-test-tickets
task run            # python loop.py --once --repo TARGET
```

Outcome line: T900 waiting round 1, still missing value, criteria

Expected `--table-only`

role	writes	scope
orchestrator	no	nothing
doer	yes	tickets/**
judge	no	nothing

If judge prints `yes`, stop. The port is wrong.

Troubleshooting

SYMPTOM	CAUSE	FIX
Judge writes yes	tools list leaked Write	strip with NO_WRITE
Writes to app/	deny envelope typo	tests assert hookSpecificOutput
Duplicate comments	marker gap	copy Deep Agents filter
SDK import error in pytest	imported at module top	lazy import, tests stub

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Validation

- [] task test green without a key
- [] table: judge writes no
- [] deny shape matches tests key by key
- [] live poll grooms T900
- [] you know the marker gap exists

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GitHub Actions

ENHANCER_BACKEND=agent-sdk

```
python3 loop.py --once --repo "$GITHUB_WORKSPACE" --ticket "$TICKET"
```

Secret ANTHROPIC_API_KEY. **Copy** labs/lab1_enhancer/workflows/enhance-on-issue.yml **onto your CRM fork.**

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Recap

The harness is the product. The framework is not.

If a port imports a shared engine, the design leaked. This folder copies `roleplan.py` and `write_scope.py` on purpose.

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